+39 393 006 3841
andreabonacci95@protonmail.com
linkedin.com/in/andrea-bonacci
github.com/AndreaBonn
andreabonn.github.io
Cagliari, Italy

Andrea *Bonacci*

AI ENGINEER — LLM & AGENTIC SYSTEMS

PROFILE

From Data Scientist to **AI Engineer**, *with* **over 4 years of experience** *in data-driven consulting for enterprise clients. I design and ship production AI systems: agents and agentic workflows with MCP and tool use, LLM integration (OpenAI, Anthropic), RAG and real-time voice AI. Solid background in end-to-end data analysis: pipelines, BI dashboards and NLP models, from ETL to deployment with FastAPI and Docker.* **Academic tutor** *for university interns and responsible for* **technical interviews** *for post-acquisition hiring. I also delivered an intensive* **8-hour university course** *on AI applied to coding.*

PROFESSIONAL EXPERIENCE

LINKALAB S.R.L. · CAGLIARI — part of Lynx Group since 2025 — Nov 2021 → present

Consulting company — Internal growth

AI Engineer & Data Analyst

Oct 2025 → present

AI Engineering focus: AI agents and agentic workflows with MCP, tool use, sub-agents and LLM integration (OpenAI, Anthropic). Production AI systems on client projects, including real-time conversational voice AI, agents for document and accounting automation, and RAG for compliance validation against regulatory corpora. Claude Code, Windsurf and Gemini CLI in production. Academic tutor for university interns. Technical interviews for hiring Data Scientists and AI Engineers post-acquisition.

`AI Agents` `MCP` `LLM Integration` `Realtime Voice AI` `RAG` `Document AI` `Claude Code` `Academic Tutor` `Technical Hiring`

Data Analyst & Data Scientist

Jul 2022 → Oct 2025

End-to-end data pipelines and BI dashboards (Power BI, QuickSight, Tableau, Looker Studio) for enterprise clients. NLP, sentiment analysis, time series. Deployment with FastAPI and Docker on AWS.

`Python` `SQL` `Power BI` `QuickSight` `Tableau` `NLP` `FastAPI` `Docker` `AWS S3/Athena`

Data Scientist — internship and contract

Nov 2021 → Jun 2022

Master's thesis internship (Italian-language Emotion Analysis for online reputation), then hired; first independent projects for enterprise clients.

Speaker — University Course

April 2026

UNIVERSITY OF CAGLIARI

Designed and delivered two intensive sessions (**8 hours total**) on *"Artificial Intelligence Applied to Coding"*: LLM architecture, prompt engineering, agentic coding with CLI tools, MCP and custom Skills.

`AI for Coding` `Prompt Engineering` `LLM` `MCP` `University Lecture`

AT A GLANCE

4+ years of professional experience

40+ open source repositories

DIFFERENTIATORS

AI + BUSINESS

Business degree + Data Science master's = I build AI that solves real problems, from idea to deployment.

AI ENGINEERING

Advanced use of LLMs, AI agents and MCP to build intelligent workflows and automate complex tasks.

GOVERNED, MEASURED AI

Agents with identity, policy and audit, built for regulated industries. RAG evaluated with quality metrics.

UNIVERSITY SPEAKER

AI applied to coding course, University of Cagliari, April 2026 — 8 hours.

MENTORING & HIRING

Academic tutor for university interns. Leading technical interviews for post-acquisition team expansion.

SKILLS

AI & LLM

AI Agents · **MCP** · **Prompt Eng.** · **RAG** · OpenAI API · Anthropic API · Tool Use · Function Calling · Document AI · Human-in-the-Loop · Realtime Voice · Sub-agents · Hugging Face · Ollama · **LLM Eval** · Agent Skills · DeepEval · Agentic Coding Config

GOVERNANCE & SECURITY

Policy-as-Code · OPA/Rego · Keycloak · Audit Trail · Least Privilege · Responsible Disclosure

CLIENT PROJECTS

ON A CONTRACT BASIS FOR ENTERPRISE CLIENTS · DETAILS ANONYMIZED FOR CONFIDENTIALITY

AI Governance Platform for Agents

2026 · in-house product

Gateway and control plane that make AI agents usable in regulated industries: every call to models and tools goes through identity, policy decision and audit.

- LLM gateway with agent identity (Keycloak), policy-as-code (OPA/Rego), append-only audit trail and EU data residency.
- Multi-tenant control plane designed and built single-handedly: human-in-the-loop approvals, policy blocks, per-agent costs (FinOps).
- Sandboxed agent with 19 MCP tools and 100 policy rules, fail-closed multi-user isolation; enterprise chat assistant in a client trial.

`AI Governance` `MCP` `Policy-as-Code` `OPA/Rego` `Keycloak` `LiteLLM` `FastAPI · React`

Conversational RAG Agent for a Manufacturer's Website

Multilingual assistant for the public website of an international industrial group, with answers that cite the source document section and a log of unanswered questions for the editorial team.

- Hybrid lexical + vector retrieval over 800+ technical documents: relevant context raised from 2% to 92%.
- Continuous evaluation with DeepEval: faithfulness 1.00, contextual recall 0.90, no undue refusals.
- AWS Bedrock in an EU region, pgvector, deployed on EC2; 4,000+ tests at 100% coverage.

`RAG` `Hybrid Search` `LLM Eval` `AWS Bedrock` `pgvector` `DeepEval`

Real-Time Conversational Voice AI Systems

Low-latency speech-to-speech AI conversation, with automated evaluation of interactions.

- OpenAI Realtime API, Gemini Live and streaming transcription (AssemblyAI), with multi-provider fallback and circuit breakers.
- RAG knowledge base queried via function calling; hallucination checks during evaluation.

`Realtime Voice AI` `OpenAI Realtime API` `Function Calling` `Circuit Breaker`

RAG & Document Intelligence for Compliance

Compliance validation of contracts, policies and playbooks against a regulatory corpus, with source citations and human review.

- Hybrid search (BM25 + embeddings) and multi-stage agentic orchestration for structured verdicts with regulatory references.
- Every finding links back to its source, to block hallucinations; human-in-the-loop with checkpoints and audit trail (async FastAPI, WebSocket).

`RAG` `LLM Orchestration` `Structured Output` `Human-in-the-Loop`

DATA SCIENCE

Python · SQL · NLP · Sentiment Analysis · Pandas · Time Series · Computer Vision

BUSINESS INTELLIGENCE

Power BI · QuickSight · Tableau · Looker Studio

MLOPS & CLOUD

Docker · FastAPI · AWS S3 · Athena · AWS Bedrock · pgvector · Flask · GitHub

LANGUAGES

Italian	Native
English	Fluent

AI Agents for Document & Finance Automation

Automation of document and accounting processes, such as reconciliation of related documents.

- Agents with function calling and validated output (Pydantic); extraction from XML, PDF and digitally signed files.
- Deterministic matching with tolerances and confidence scoring, ERP integration and human review of uncertain cases.

`AI Agent` `Document AI` `Information Extraction`

PERSONAL PROJECTS & OPEN SOURCE

GITHUB.COM/ANDREABONN · 40+ PUBLIC REPOSITORIES

Local AI: Voice, Transcription, Images

Ongoing

Clone My Voice: offline voice cloning and dubbing on GPU, 6 isolated TTS engines, Whisper, blind A/B tests. Speech-to-Text: from recorded lectures to study material with faster-whisper, a local LLM and evaluated hybrid RAG (semantic retrieval 100% vs 56% BM25 on 82 questions). Image Generator on SDXL and FLUX, video anonymization with YOLO.

`Voice Cloning` `Whisper` `RAG Eval` `Diffusion` `Ollama` `CUDA`

Agentic Tooling

Ongoing

Claude Phone: Claude Code driven from Telegram, with a permission gate on tools (MCP included), human approval and audit log. Two GitHub Marketplace Actions with multi-provider LLM fallback: AI PR Reviewer and AI Changelog Generator. Claude Code Tools Explorer (TUI) and multi-account manager for Antigravity CLI.

`Claude Code` `Human-in-the-Loop` `GitHub Actions` `Textual`

Web & Desktop Products

Ongoing

10 live PWAs with end-to-end ownership, including Learn Italian (150 lessons A1-C2, spaced repetition), Meteo & Spiagge, PlotMark, BetTracker (OCR with Gemini) and apps for personal finance, health and shared expenses. Linux desktop tools: remote shell via Telegram, Command Quiver, Sysbar, Nautilus extensions.

`React` `TypeScript` `Firebase` `PWA` `GTK4`

Security & Responsible Disclosure

Aug 2026

Reported a critical RCE vulnerability (unauthenticated code execution as SYSTEM) in WinBoat, an open source project. Private responsible disclosure with runtime-verified patches, fixed within 4 hours of first contact. Credited in the v0.9.1 security release.

`Responsible Disclosure` `RCE` `Open Source`

EDUCATION

Master's Degree — Data Science & Business Informatics

2022

University of Pisa

Thesis: Italian Emotion Analysis for online reputation monitoring

Bachelor's Degree — Business Administration

2018

University of Pisa